

Project: BladeCore

M54C Evaluation Board

Rev 1.0.0

21-03-2026

CONTENT

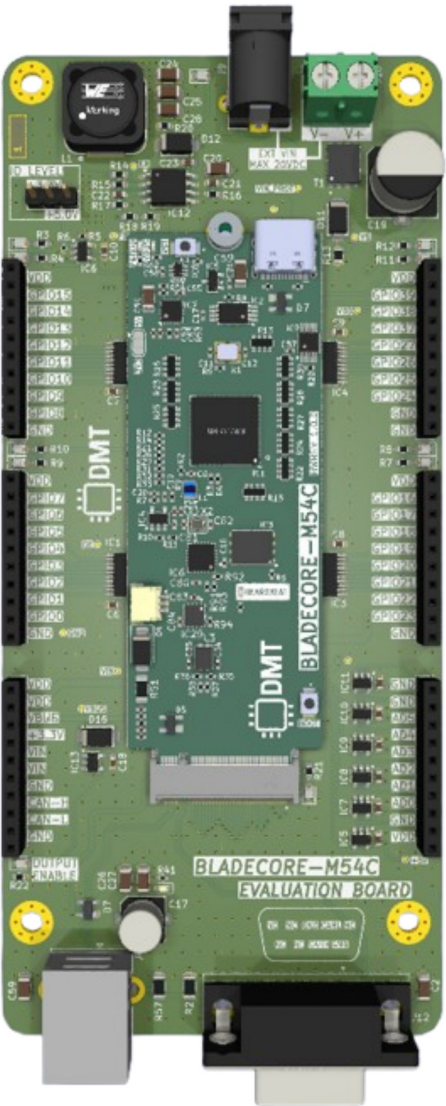
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COMMENT GUIDELINES

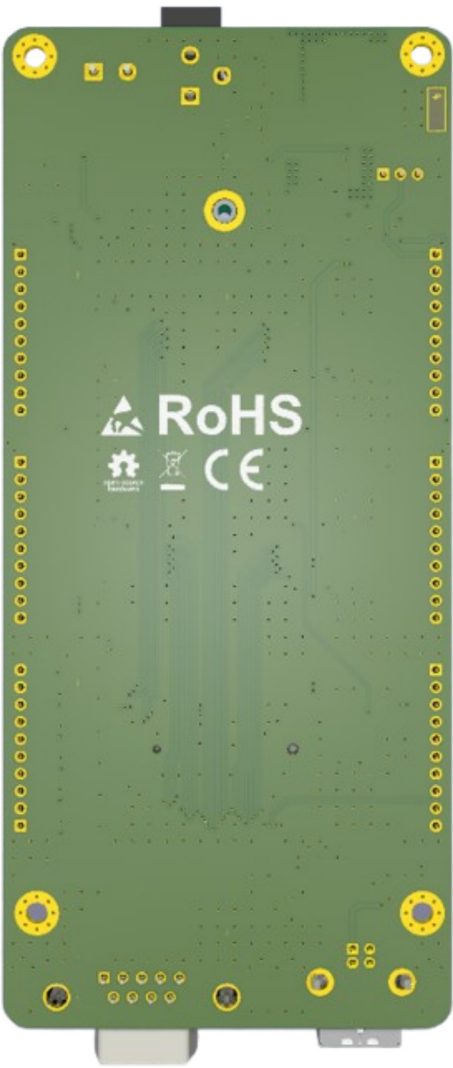
General comments are black, 50 mil size
Design notes and guidelines are blue, 50 mil size
Layout instructions are red, 50 mil size

PCB PREVIEW

TOP VIEW



BOTTOM VIEW

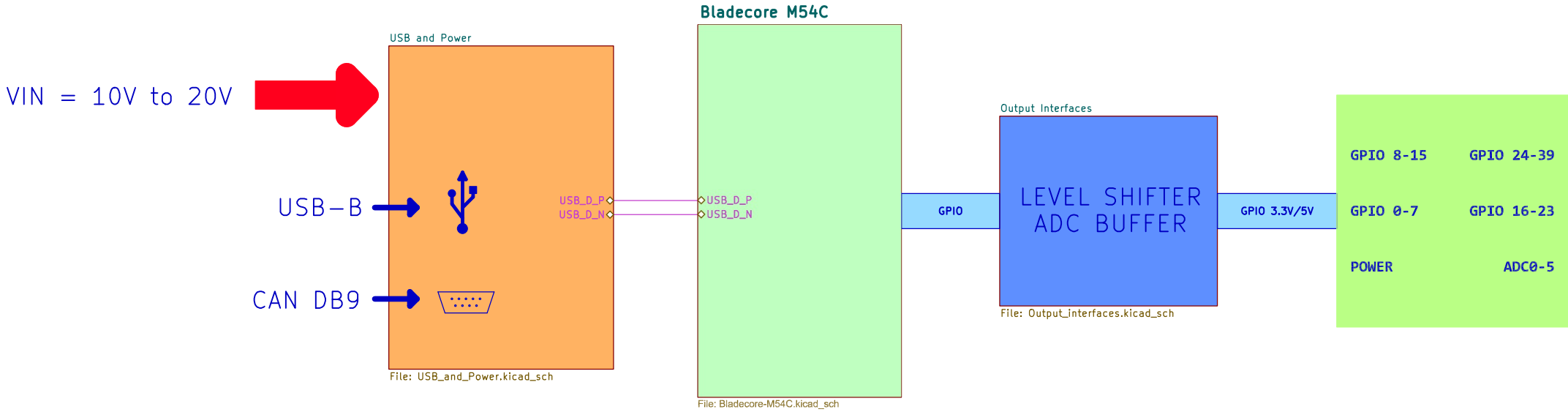


NOTES

Not fitted components are marked as X

 Date: 2025-01-12 Sheet Title: Root	Board Name: BladeCore		Project Name: M54C Evaluation Board	
	File Name: M54-DevBoard_1.0.0.kicad_sch		Revision: 1.0.0	Variant:
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 1 of 6

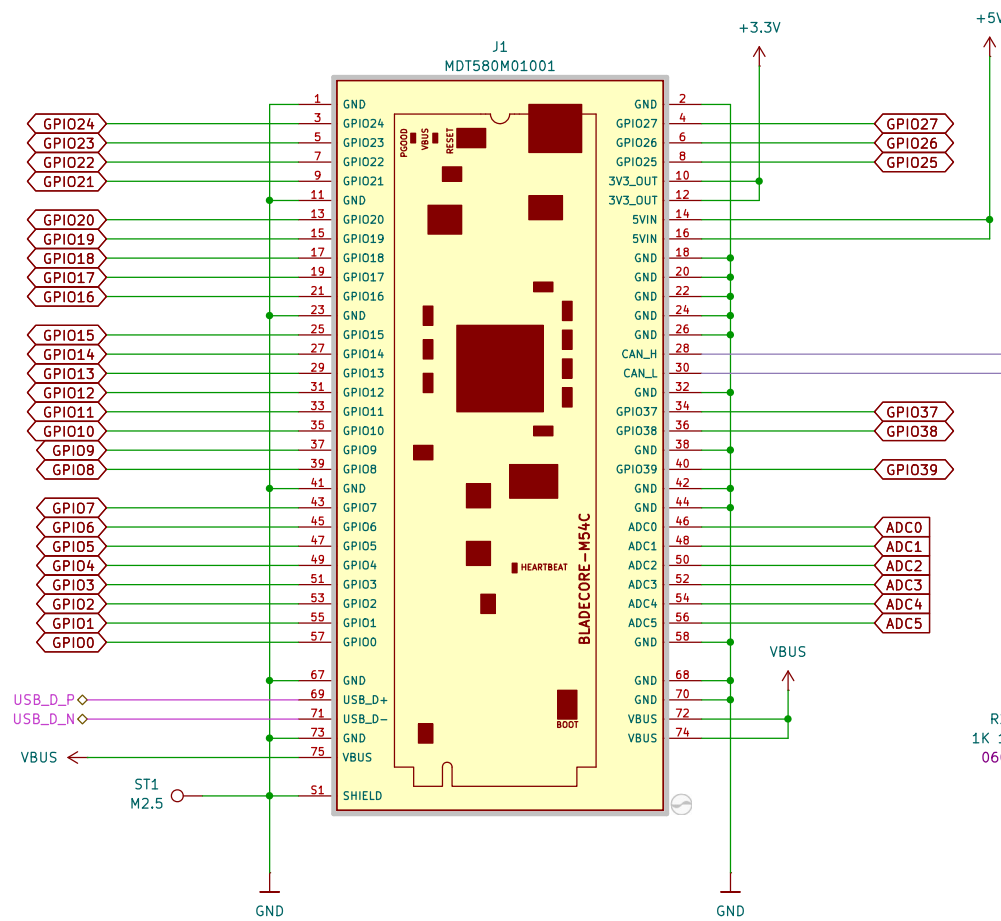
[3] Project Architecture



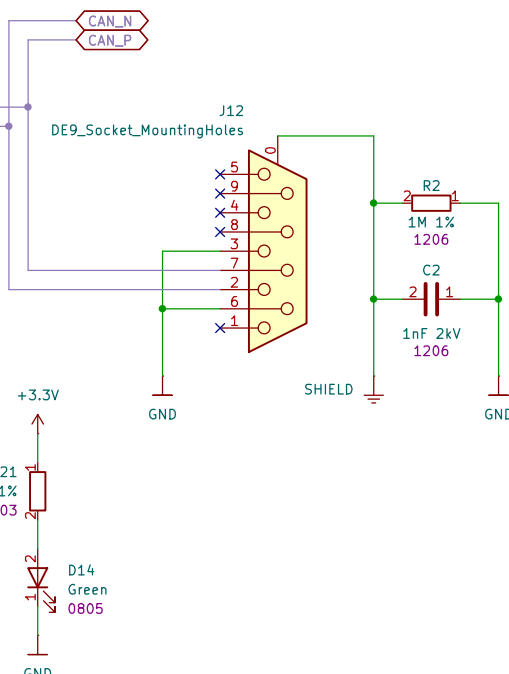
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	File Name: Project Architecture.kicad_sch		Revision: 1.0.0	Variant:
	Sheet Title: Project Architecture	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3
				Sheet: 3 of 6

[4] Sheet Title A

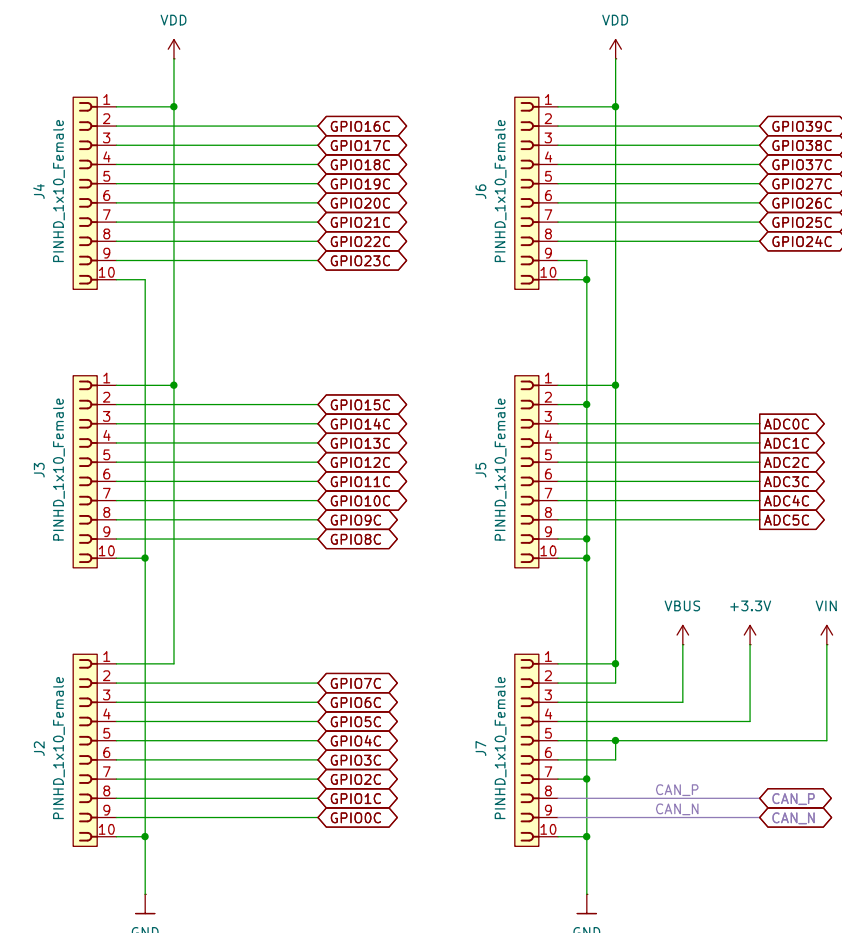
BLADECORE-M54C SOCKET



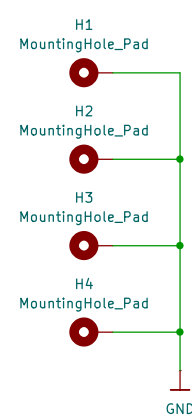
CAN DB9 CONNECTOR



GPIO HEADERS

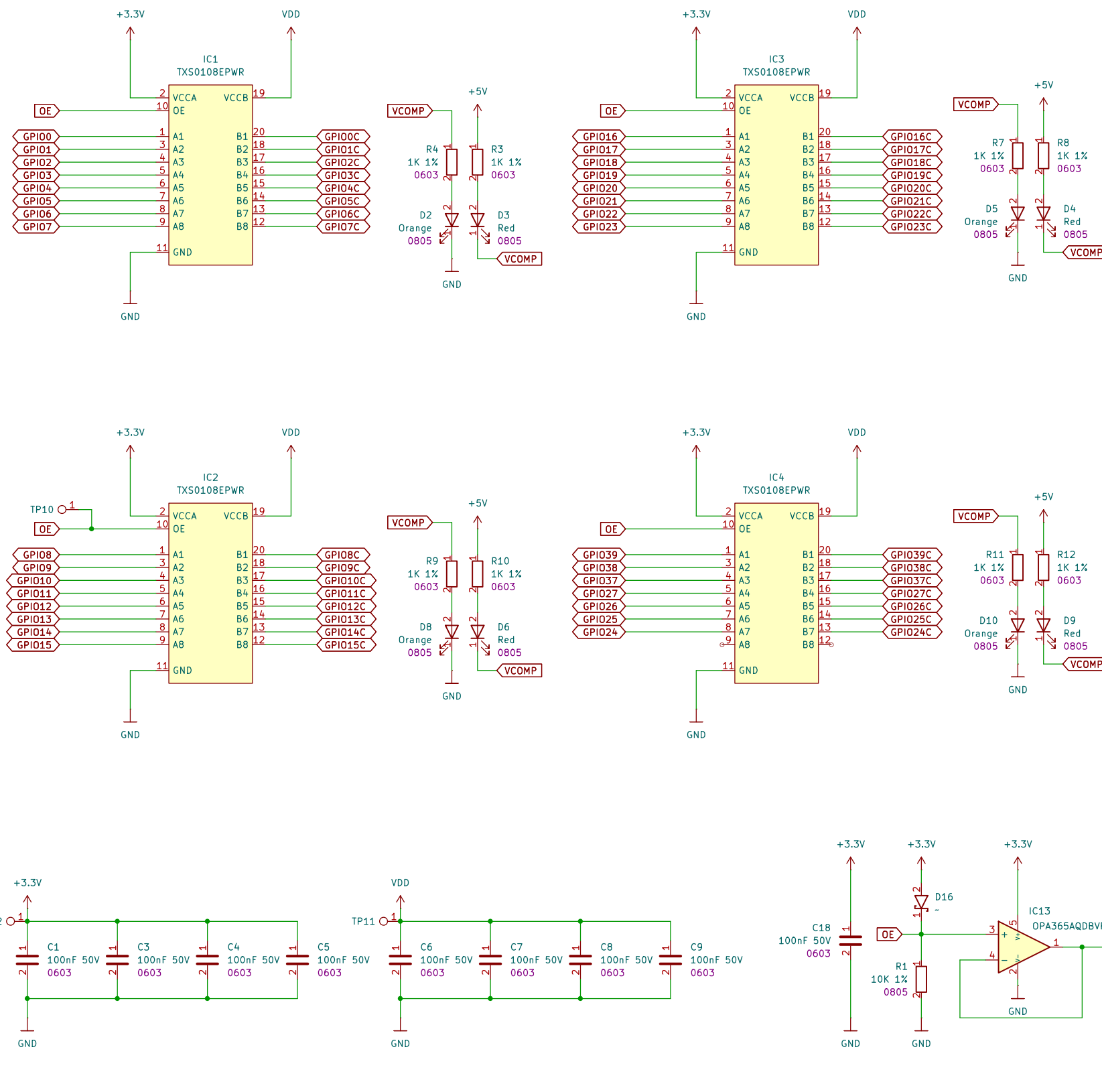


MOUNTING HOLES

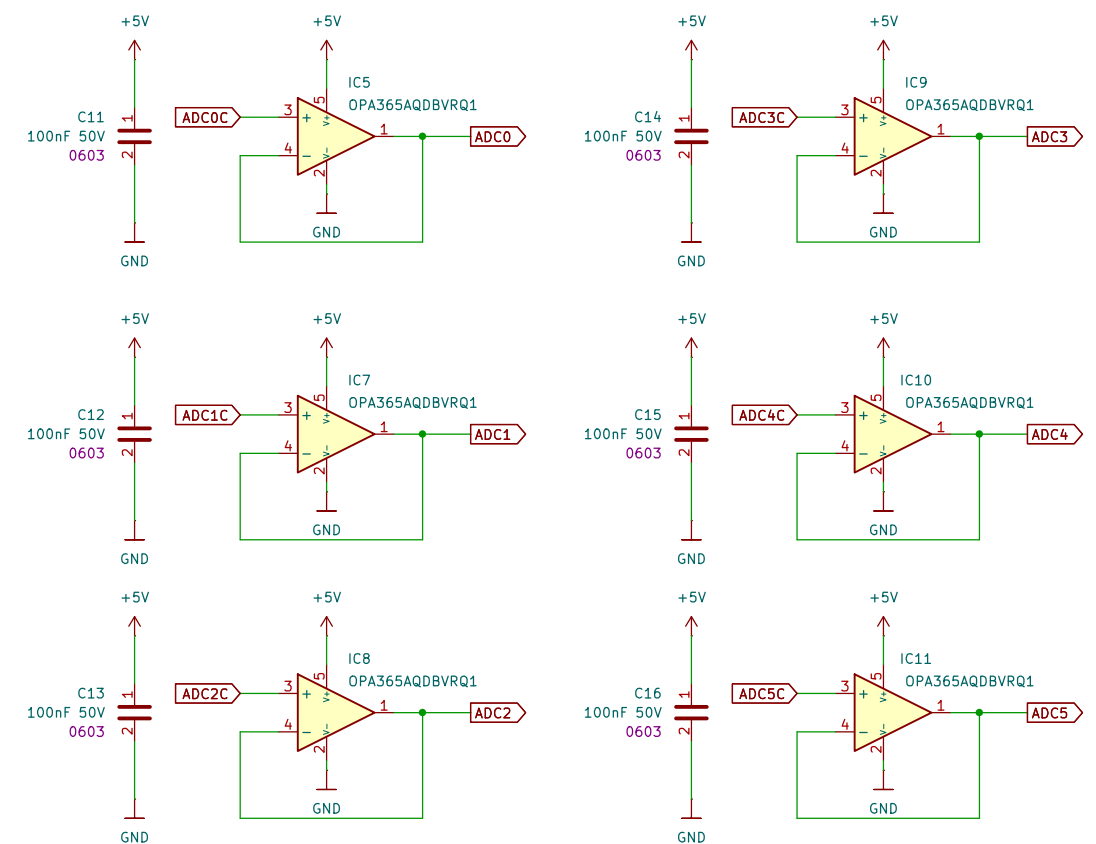


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	File Name: Bladecore-M54C.kicad_sch		Revision: 1.0.0	Variant:
Sheet Title: Bladecore M54C	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 4 of 6

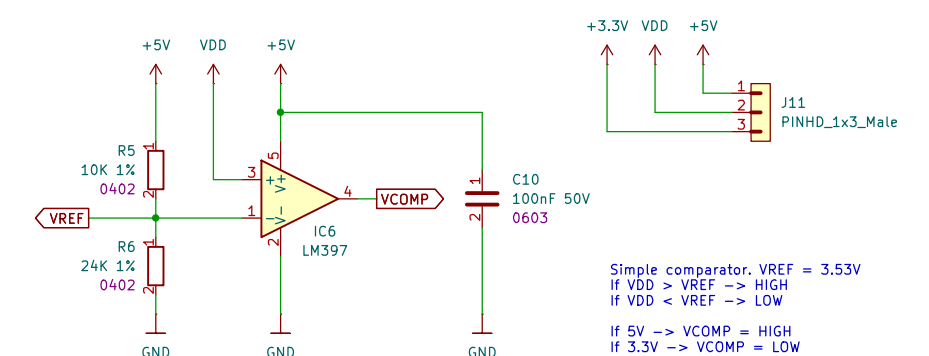
GPIO LEVEL SHIFTERS



ADC INPUT BUFFERS

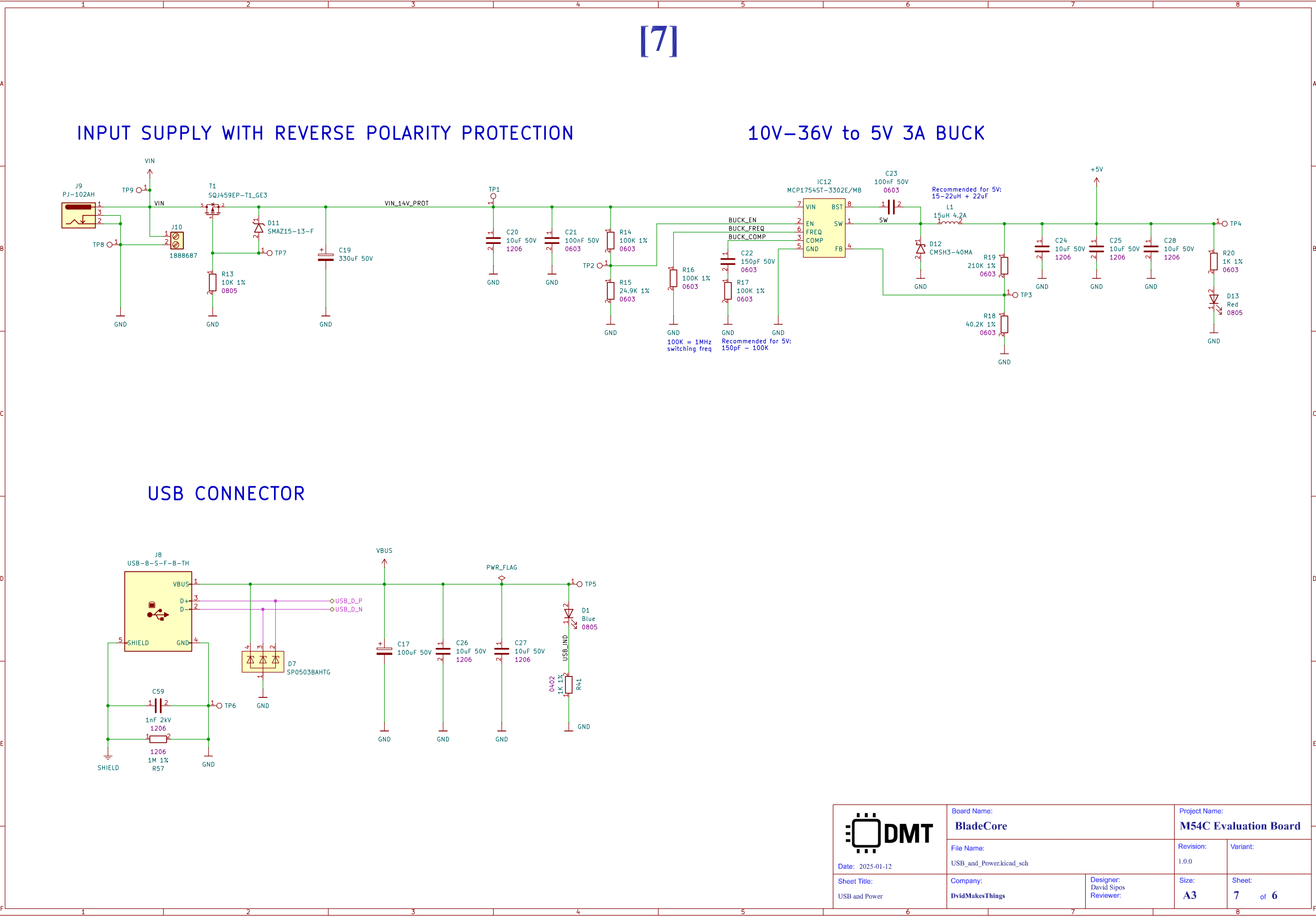


VDD LEVEL SELECTOR AND COMPARATOR



Simple comparator. $V_{REF} = 3.53V$
 If $V_{DD} > V_{REF} \rightarrow \text{HIGH}$
 If $V_{DD} < V_{REF} \rightarrow \text{LOW}$

If $5V \rightarrow V_{COMP} = \text{HIGH}$
 If $3.3V \rightarrow V_{COMP} = \text{LOW}$

[illegible][illegible][illegible]

1	2	3	4	5	6	7	8
[20] Revision History							
A							
B							
C							
D							
E							
F	1	2	3	4	5	6	8

 Date: 2025-01-12	Board Name: BladeCore		Project Name: M54C Evaluation Board	
	File Name: Revision History.kicad_sch		Revision: 1.0.0	Variant:
	Sheet Title: Revision History	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3 Sheet: 20 of 6